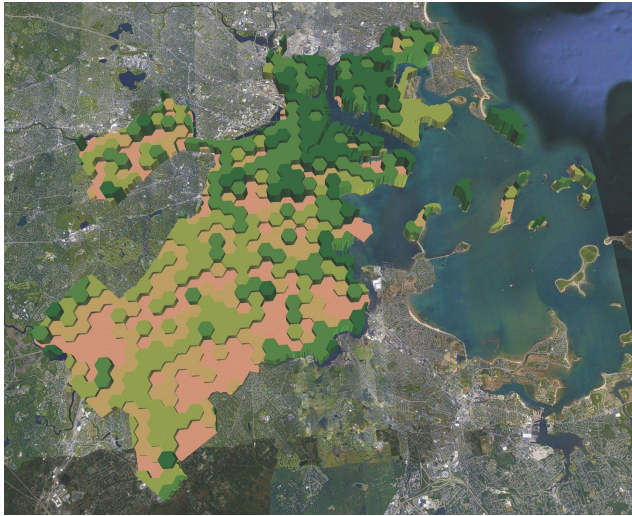




2014-2019 Canopy Change Assessment

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New to GIS and want to use this data? Or want more information about the project and an overview of the available data? Check out this [visual guide](#).

Fun fact: Shapefiles and rasters can be downloaded and then displayed in Google Earth Pro! Instructions for viewing data in Google Earth Pro can be found [here](#).

Left: screenshot of canopy change data in Google Earth Pro with colors and heights of hexagons corresponding to numerical values.

Overview

A number of Canopy Change Assessment 2014-2019 datasets are available for download. This includes:

- Tree canopy coverage and change analysis summarized by geographies
- Tree canopy tops
- Tree canopy height
- Land use by parcels
- Land cover
- Tree tops showing canopy growth, loss, no change.

A note on East Boston and Boston Logan International Airport: The presence of Boston Logan International Airport within East Boston's neighborhood boundaries is a unique condition. This project takes the position that the airport is as much a part of the city as anything else with impacts on the surrounding area and so the statistics in the report and nearly all shapefiles include the airport. However, we received questions about the airport's effect on canopy coverage statistics after the 2014 analysis. In response, we ran a city and neighborhood-level canopy coverage analysis to see what statistics looked like when including and then excluding the airport. Those results can be viewed in the city and neighborhood geographies.

Projection

NAD 1983 StatePlane Massachusetts FIPS 2001 (US Feet)_1

Map units

US survey feet

Credits

Analysis commissioned by the Boston Parks and Recreation Department and conducted by the University of Vermont Spatial Analysis Laboratory.

Use limitations

There are no restrictions on the use of this dataset. However, the City of Boston and the University of Vermont make no representations of any kind, including but not limited to the warranties of merchantability or fitness for a particular use, nor are any such warranties to be implied with respect to the data.

Geographies

Neighborhood (NBHD)

Neighborhood boundaries come from the 2015-2021 Open Space and Recreation Plan. Please see the [earlier note](#) in the Overview section describing the assessment's approach to Boston Logan International Airport.

Parcel

Fiscal Year 2019 (July 1, 2018- June 30, 2019) parcel boundaries and information available for download through Analyze Boston. Created by the Assessing Department.

Hexagon (Hex)

25 hectare hexagons uniquely created for this project.

Ward

City of Boston wards as of 2019.

Census tracts (CT)

2010 census tract geographies.

Open space protected

Based on the Open Space shapefile as of 2019. Land displayed as "open space protected" were selected by choosing properties marked as "X" under the POS column in the Open Space shapefile.

City

City boundary derived from BPRD neighborhood boundaries and used for all analyses in this assessment. Please see the [earlier note](#) in the Overview section describing the assessment's approach to Boston Logan International Airport.

Shapefiles**Tree canopy metrics** (applies to all geographies)

Tree Canopy (TC) Assessment metrics for Boston, MA. These datasets consist of TC metrics summarized using various geographies. The metrics presented in these tables are based on 2019 high resolution land cover and 2019 leaf-on LiDAR. The TC Assessment is a top-down approach to analyzing the forest. Its purpose is to integrate high resolution land cover data with other GIS datasets to produce a set of detailed metrics on the forest that allow decision makers to know how much tree canopy currently exists (termed Existing TC). Existing TC is determined by extracting all features classified as tree canopy from a high resolution land cover dataset. Possible TC is queried out from this overlay and consists of all land that was not existing canopy, not water, not a building, and not a road. Possible TC is further divided into two subcategories: Possible-impervious and Possible-vegetation. Possible-impervious consists of all impervious land that, through modification, could support tree canopy. Examples of such features are parking lots, driveways (through overhanging coverage) and playgrounds. Possible-vegetation consists of all land that is low-lying vegetation, primarily grass or shrubs, which could conceivably be converted to support tree canopy. Examples of such features include residential lawns and playing fields. TC metrics do not serve to address the issues of where it is socially desirable or financially feasible to plant trees. Rather, the TC metrics serve as the basis for

beginning to form answers to these questions. TC metrics are presented in the attribute table as both absolute area (in map units) and relative area (percentage of land area) per parcel. For example, an Existing TC Area (TC_E_A) value of 13,677 and an Existing TC Percentage (TC_E_P) of 21.8 indicate that for the area in question the area of Existing TC is 13,677 (**in map units**) and 21.8% of that feature is tree canopy.

TC_E_A = Tree canopy existing area. The area of tree canopy present when viewed from above using aerial or satellite imagery, excluding water.

TC_PV_A = Possible vegetation area. Grass or shrub area that is theoretically available for the establishment of tree canopy.

TC_Land_A = Land area. Land area excluding water bodies.

TC_Pi_A = Possible impervious area. Asphalt or concrete surfaces or bare soil, excluding roads and buildings, that are theoretically available for the establishment of tree canopy.

TC_P_A = Possible area. Area theoretically available for establishment of tree canopy.

TC_E_P = Existing percent. The amount of tree canopy present when viewed from above using aerial or satellite imagery, excluding water as a percentage. This is the value most used for reporting out current canopy coverage.

Can_P = Canopy as a percent of all land cover (including water).

TC_Pv_P = Possible vegetation percent. Grass/shrub area that is theoretically available for the establishment of tree canopy as a percentage of land area.

TC_P_P = Possible tree canopy percent. Area theoretically suitable for tree canopy as a percentage of land area.

TC_Pi_P = Possible impervious canopy percent. Asphalt or concrete surfaces or bare soil, excluding roads and buildings, that are theoretically available for the establishment of tree canopy as a percentage of land area.

Tree canopy change metrics (applies to all geographies)

Tree Canopy Change Assessment metrics for Boston, MA. These datasets consist of tree canopy change metrics summarized using various geographies. The metrics presented in these tables are based on 2019 leaf-on LiDAR, 2019 high resolution land cover, and the high resolution 2019 tree canopy layer (Tree Canopy Change 2014-2019.tif). The Tree Canopy Change Assessment is a top-down approach to analyzing forest change for over the period of 2014 and 2019. Its purpose is to integrate high resolution land cover data with other GIS datasets to produce a set of detailed metrics on the forest that allow decision makers to know how much tree canopy was gained (termed Gain), how much tree canopy was lost (termed Loss), and how much tree canopy did not change over the given time period (termed No_Change). Existing TC is determined by extracting all features classified as tree canopy from a high resolution land cover dataset. The area in which there has been a

change in tree canopy (termed Change_Are) is calculated using the TreeCano_1 (2019) subtracted by the TreeCanopy (2014), if this value is negative then this represents a loss in tree canopy over the time frame given. The percent of tree canopy change (termed Change_P1) is calculated using TreeCano_3 (2019) subtracted by TreeCano_2 (2014), if this value is negative then this represents a loss in tree canopy over the time frame given.

LandArea= Area excluding water bodies

Gain= Area of canopy gain between the two years

Loss= Area of canopy loss between the two years

No_Change= Area of no canopy change between the two years

TreeCanopy= 2014 total canopy area (baseline)

TreeCano_1 = 2019 total canopy area

Change_Are= The change in area of tree canopy between the two years

Change_Per= Relative change calculation used in economics is the gain or loss of tree canopy relative to the earlier time period: $(2019 \text{ Canopy} - 2014 \text{ Canopy}) / (2014 \text{ Canopy})$

TreeCano_2 = 2014 canopy percentage

TreeCano_3= 2019 canopy percentage

Change_P_1= Absolute change. Magnitude of change in percent tree canopy from 2014 to 2019 $(\% 2019 \text{ Canopy} - \% 2014 \text{ Canopy})$

Heat metrics (applies to census tract and hexagon geographies)

These datasets consist of summer temperature metrics for Boston, MA. These heat metrics summarize six CAPA Urban Heat Watch program temperature and heat index datasets using geographical boundaries from the Hexagons (HEX) layer and the census tracts (CT) layer. Heat datasets were created by Museum of Science, Boston, and the Helmuth Lab at Northeastern University. Heat metrics are presented in the attribute table as mean values of each Heat Watch program dataset for all hexagon and census tract features. The six heat values included in these tables are July 2019 temperature and heat index in degrees Fahrenheit for each of 3 1-hour periods -- 6 a.m., 3 p.m., and 7 p.m. EDT. The geographic boundaries used to summarize the heat metrics are current as of 2019.

Mean_am_T_= mean morning temperature within geography

Mean_am_HI= mean heat index temperature within geography

Mean_p2_T_= mean temperature of all times within geography

Mean_p2_HI= mean heat index of all times within geography

Mean_ev_T= mean evening (7pm) temperature within geography

Mean_ev_HI= mean evening (7pm) heat index within geography

Tree canopy height change (Tree Canopy Height Change 2014-2019)

Change in tree canopy heights between 2014 and 2019. Derived from satellite imagery and leaf-on LiDAR.

Tree canopy polygons (2019 Tree Canopy Polygons)

2019 tree canopy polygons derived from satellite imagery and leaf-on LiDAR.

Land use (Parcels FY19 Land Use)

Layer derived from Fiscal Year 2019 assessing parcels.

Project generalized the LU codes in their FY 2019 parcels dataset to create a new attribute LU_general as shown below. Open space was added as a land use category by overlaying Boston's Open_space dataset onto the parcels (open space taking precedence over underlying parcel land use designation). Dissolve on generalized LU codes. Overlay with city boundary (BostonCity_AOI) to identify ROW (anything not covered by parcels or open space polygons).

FY19 parcels label	LU_general
-	Unknown
A	Residential
AH	Commercial
C	Commercial
CL	Commercial
CM	Residential
CP	Residential
E	Institutional
EA	Institutional
I	Industrial
R1	Residential
R2	Residential
R3	Residential
R4	Residential
RC	Mixed Use
RL	Residential

Rasters**Tree canopy change** (Tree Canopy Change 2014-2019)

Tree-level raster image depicting canopy growth, loss, or no change.

Value: 1= No change, 2= Gain, 3= Loss

Count: pixel count

Land cover metrics (Land cover 2019)

This dataset consists of land cover metrics for Boston, MA. The land cover metrics summarize the 2019 land cover dataset (landcover_2019_boston.tif) using various geographical boundaries. Land cover metrics are presented in the attribute table as both absolute area (in map units) and relative area (percentage of land area) of each land cover class for all features. The land cover classes included in this table are 1) Tree Canopy, 2) Grass/Shrubs, 3) Bare Soil, 4) Water, 5) Building, 6) Road/Railroad, and 7) Other Paved Surface. In addition, the table includes metrics for pervious and impervious surfaces. Pervious surfaces include tree canopy, grass/shrubs, bare soil, and water. Impervious surfaces include buildings, roads, and other paved surfaces. The land cover dataset used to derive these metrics is primarily based on 2018 NAIP, 2019 Orthos, and 2019 LiDAR data. The geographic boundaries used to summarize the land cover metrics are current as of 2019.